

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

Department of Computer Science & Engineering

Capstone Project-I (CS4095) | Autumn Semester 2026-27

RESEARCH PLAN

Risk-Aware Deep Reinforcement Learning for Building Energy Optimisation under Uncertain Demand, Solar Generation, and Time-of-Day Tariffs

A forecast-informed, comfort-constrained control framework for grid-interactive buildings

Name of Candidate	Prajwal Acharya
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Supervisor(s)	Dr. Sibarama Panigrahi
Broad Research Area	Deep Learning, Reinforcement Learning, Sustainable Computing and Smart Energy Systems
Research Period	August 2026 - November 2026

Submitted in accordance with the Instruction Sheet for Submission of Research Plan for B.Tech CP-I.

1. Student Details

Candidate	Prajwal Acharya
Roll Number	123C50143
Supervisor(s)	Dr. Sibarama Panigrahi
Tentative Project Title	Risk-Aware Deep Reinforcement Learning for Building Energy Optimisation under Uncertain Demand, Solar Generation, and Time-of-Day Tariffs

2. Brief Problem Statement

2.1 Research Problem

Buildings increasingly combine air-conditioning systems, rooftop solar photovoltaic generation, batteries, electric-vehicle charging and other flexible loads. Yet their operation is usually reactive: air-conditioning follows a fixed thermostat, batteries are charged or discharged without anticipating solar availability, and flexible consumption often overlaps with expensive peak-tariff periods. This increases electricity cost, peak grid demand and carbon impact, while indiscriminate optimisation can reduce occupant comfort or stress equipment.

The proposed research addresses sequential energy-control decisions in a grid-interactive building. At every control interval, the controller observes the building state and forecasts of demand, weather, solar generation and tariff. It then chooses safe actions - initially HVAC temperature-setpoint adjustment and battery charging/discharging - that minimise operating cost and peak demand without violating comfort, battery or grid limits.

2.2 Motivation and Relevance

Electricity demand from cooling and distributed energy resources is becoming more variable. Dynamic and Time-of-Day tariffs create an opportunity to reduce bills by shifting flexible loads away from peak periods and towards periods of rooftop solar availability. However, a useful controller must make decisions before the peak occurs. This requires learning from multivariate time-series data and reasoning about the consequences of actions over several hours.

The project is relevant to hostels, campuses, offices and residential complexes where HVAC operation and storage usage affect both user bills and grid peaks. It is also directly relevant to India-oriented operating conditions: hot-weather cooling demand, variable solar output, time-varying tariffs and potentially intermittent occupancy. The outcome will be tested first in a simulator, which provides a safe and reproducible setting before any real-world actuation is considered.

2.3 Research Gap

Deep learning has achieved strong short-term load and solar forecasts, while deep reinforcement learning (DRL) has been used for HVAC and storage control. Nevertheless, much existing work has one or more of the following limitations:

- It optimises using point forecasts, despite forecast error being largest during unusual weather, occupancy changes and solar variability.
- It encodes safety only as a soft reward penalty, which does not reliably prevent comfort, battery or grid-limit violations.
- It reports average cost or energy savings without evaluating failure behaviour under biased forecasts, heatwave conditions or tariff changes.
- It compares RL only with a simple thermostat rather than separating the benefits of forecasts, uncertainty modelling and safety constraints.

This project will examine whether calibrated uncertainty information can make a DRL energy controller more reliable. Rather than being aggressive whenever a forecast predicts savings, the controller will adapt its risk level and fall back to safe actions when uncertainty is high.

2.4 Proposed Approach and Research Questions

The proposed system combines a probabilistic forecasting module, a DRL controller and a safety shield. The forecasting module estimates near-future building demand and solar generation together with prediction intervals. The controller receives both the forecasts and their uncertainty, and learns a policy using PPO as the main initial algorithm. The safety shield projects or replaces unsafe actions to maintain operational constraints.

RQ1: Can forecast uncertainty improve the cost-comfort trade-off compared with a controller that uses only point forecasts?

RQ2: Can a safety shield reduce constraint violations during forecast errors and high-demand events with only a limited loss of average cost saving?

RQ3: How robust is the proposed controller across changed tariffs, solar availability, occupancy patterns and weather conditions?

3. Objectives for Autumn Semester 2026-27

The semester work is intentionally limited to a simulation-based prototype and rigorous evaluation. The objectives are:

- Review recent work on building energy management, probabilistic time-series forecasting, safe reinforcement learning and demand-response optimisation.
- Formulate a Constrained Markov Decision Process (CMDP) for a building with HVAC and battery flexibility, including state variables, action bounds, reward and safety constraints.
- Set up a reproducible CityLearn-based energy-management environment with demand, weather, solar PV, tariff and storage data.
- Develop baseline controllers: fixed-schedule/rule-based control and a standard PPO controller without uncertainty-aware inputs.
- Develop a probabilistic demand/solar forecasting module using quantile regression or an LSTM/GRU-based model, and assess calibration as well as point accuracy.
- Integrate forecast intervals into the RL state and build a safety shield or fallback mechanism for comfort, battery state-of-charge and grid-import limits.
- Conduct ablation and robustness experiments under forecast noise, tariff changes, high-demand conditions and changed solar profiles.
- Prepare a technical report/manuscript and a compact dashboard or demonstration that explains recommended actions and expected savings.

4. Detailed Month-wise Work Plan (August-November 2026)

Month	Activities	Output / Evaluation Evidence
August	Literature review; define research questions; formulate state, action, reward and constraints; install and explore CityLearn; implement a fixed-schedule and tariff-aware rule-based baseline.	Literature matrix; mathematical problem formulation; reproducible simulator configuration; baseline result plots.
September	Prepare time-series features; build and evaluate initial load/PV forecasting model; train standard PPO controller; define measurement protocol and random seeds.	Forecasting metrics (MAE/RMSE and interval calibration); PPO learning curve; initial baseline comparison.
October	Add forecast uncertainty to controller state; implement safety shield/fallback; run controlled experiments with forecast noise, tariff changes and peak-demand events.	Proposed-controller prototype; constraint-violation analysis; ablation results.
November	Robustness/generalisation testing; statistical aggregation across seeds; finalise results and limitations; prepare demonstration dashboard and research-paper draft.	Final evaluation tables/figures; working demo; complete report/manuscript draft.

5. Expected Deliverables by November 2026

- **Mathematical model:** A documented CMDP formulation with reward components, action bounds and explicit comfort, storage and grid constraints.²
- **Simulation prototype:** A reproducible CityLearn configuration and source code for rule-based, standard DRL and proposed risk-aware DRL controllers.
- **Forecasting component:** A multivariate demand/solar forecaster with point and interval-quality evaluation.
- **Experimental results:** Cost, peak demand, comfort, constraint-violation and robustness comparisons across defined test scenarios.
- **Research output:** A conference-style paper draft or technical report under preparation, including literature review, method, experiments, findings and limitations.
- **Demonstration:** A lightweight visual dashboard showing forecasts, action recommendations, demand profile, comfort status and savings comparison.

6. Milestones and Evaluation Points

Evaluation Point	Target Date	Milestone	Evidence
Mid-semester milestone	30 September 2026	Research formulation, simulator environment, baseline controllers and initial forecast model completed.	Code repository; formulation note; preliminary plots and discussion with supervisor.
Final milestone	Before the November 2026 final examination	Risk-aware safe RL prototype evaluated under defined uncertainty and stress scenarios; report and demo completed.	Final tables/figures; reproducible experiments; manuscript/report draft; dashboard/demo.

Evaluation Methodology

The proposed controller will be evaluated against fixed-schedule, tariff-aware rule-based and standard PPO baselines. Each method will be tested across multiple random seeds and scenarios: normal versus high cooling demand, accurate versus noisy forecasts, low versus high solar output and alternative tariff schedules. Primary metrics will be electricity cost, peak demand, comfort-violation duration, battery/grid constraint violations, solar self-consumption and performance under the worst forecast-error cases. The contribution will be assessed through ablation: remove uncertainty input, remove the safety shield and compare the resulting behaviour.

7. Scope, Feasibility and Risk Management

The work is feasible within one semester because it begins with a single-building simulator and only two controllable assets: HVAC setpoint and battery power. EV charging, multi-building coordination and real hardware integration are deliberately kept outside the required CP-I scope. They may be considered only after the core controller and evaluation pipeline are stable.

Risk	Mitigation
RL training instability or long runtime	Start with PPO, a reduced simulation scenario, normalised observations/rewards and fixed seeds; use rule-based baselines throughout.
Forecast data/model is insufficient	Use simulator-supported demand/solar signals and compare a simple persistence forecast, GRU/LSTM and quantile baseline before coupling to RL.
Scope becomes too broad	Restrict compulsory work to HVAC + battery; treat EV, multi-building coordination and physical deployment as future work.
Results show only small average saving	Report robust performance and reduced violation risk under adverse scenarios, not only a large average-savings claim.

8. Supervisor's Recommendation

I recommend the proposed research plan for implementation during the Autumn Semester 2026-27. The plan is realistic, well-structured and appropriate for the Capstone Project-I research period. The candidate will work under my/our supervision and report progress at the stated milestone points.

Signature of Supervisor

Name: Prajwal Acharya
Date: 11/08/2026

Prajwal Acharya
Signature of Candidate

Prajwal Acharya
Date: 11/08/2026

References

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